PROJECT PROPOSAL

RideTrack

A Smart, Cashless and Punctual Public Transport Platform for Sri Lanka

Submitted for consideration under the Clean Sri Lanka Programme

Subject	Proposal to modernise public bus and rail passenger services in Sri Lanka through live vehicle tracking, digital QR ticketing and cashless payments (RideTrack)
Submitted to	The Chairman / Secretary, Presidential Task Force for the Clean Sri Lanka Programme
Copy to	Ministry of Transport and Highways; National Transport Commission (NTC); Sri Lanka Transport Board (SLTB); Sri Lanka Railways
Submitted by	**SILVA D S J** Software Developer, RideTrack project
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Date	14 October 2026

"A cleaner Sri Lanka is also a more disciplined, transparent and punctual one. Public transport is where most citizens feel that change every day."

SILVA D S J
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14 October 2026

The Chairman / Secretary,
Presidential Task Force for the Clean Sri Lanka Programme,
Presidential Secretariat, Colombo 01.

Dear Sir / Madam,

Subject: Proposal to modernise public bus and rail passenger services through RideTrack, a live tracking, digital ticketing and cashless payment platform

I respectfully submit this proposal for your consideration under the Clean Sri Lanka Programme. RideTrack is a software platform I have designed and built that lets passengers see where their bus or train is in real time, buy tickets with a QR code instead of cash, and receive delay alerts, while giving transport authorities a live dashboard of every vehicle, trip and ticket.

The programme has made the quality, discipline and standards of passenger transport a national priority, and the Ministry of Transport has spoken publicly about fleet modernisation and a digital, cashless payment system. RideTrack can support those goals: the backend is already working, tested and deployable, and has been proven on real open timetable data. I am requesting the opportunity to present the system to the Task Force and to run a small, low-cost pilot on selected routes in Colombo together with the NTC and SLTB.

I would be grateful for the chance to discuss this proposal at your convenience.

Yours faithfully,

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1. Executive Summary

Every day, millions of Sri Lankans wait at bus halts and stations without knowing when the next vehicle will arrive, pay fares in cash with no receipt, and get no warning when a service is late or cancelled. Authorities, for their part, have little real-time visibility of how vehicles are running. **RideTrack** addresses these problems with one connected platform for three groups of users:

User	What RideTrack gives them
Passengers	Routes, stops and timetables; live vehicle positions and arrival times; QR tickets bought by card or mobile payment; delay and cancellation alerts.
Conductors / staff	Scan and validate QR tickets on a phone; report passenger occupancy.
Authorities (NTC, SLTB, Railways)	Live operations dashboard, delay detection, published alerts, ticket and revenue reports, and a web admin panel to manage routes, vehicles and trips.

The platform supports the Clean Sri Lanka goals of **discipline, transparency and service quality**: cashless tickets reduce fare leakage and disputes, live tracking encourages punctuality, and more reliable public transport draws people out of private vehicles, easing congestion and emissions. This proposal asks for official support to run a pilot (Section 7).

2. Background and Alignment with Clean Sri Lanka

The Clean Sri Lanka Programme, launched on 1 January 2025 and led by a Presidential Task Force, seeks a social, environmental and ethical renewal of the country. Public transport has been one of its most visible areas: setting quality and safety standards for passenger buses and school vans, renovating major bus stands, importing new low-floor buses for Colombo, and, as announced by the Ministry of Transport in 2025, plans for a digital cashless payment system for passengers.

RideTrack supports these goals directly:

Clean Sri Lanka goal	How RideTrack contributes
Discipline and punctuality	GPS tracking of every trip against its timetable; automatic delay detection when a trip runs 10+ minutes late; reports by route.
Transparency and ethics	Every fare is a digital, traceable transaction with a signed QR ticket; less cash handling means less fare leakage and fewer disputes.
Quality of service	Passengers know when their bus or train will come and are told about delays and cancellations straight away.
Environment	Reliable public transport encourages a move away from private vehicles; paper tickets are replaced by digital ones.
Inclusion	Live information helps elderly passengers, people with disabilities and women travelling at night spend less time waiting at halts.

3. Problem Statement

- **No real-time information.** Passengers cannot see where a bus or train is, so they wait at halts for long and uncertain periods.
- **Cash-based fares.** Cash fares are slow, lead to disputes over change and balance tickets, give passengers no proof of payment, and make revenue hard to audit.

- **Little operational visibility.** Regulators and operators lack live data on vehicle positions, delays and occupancy, which makes it hard to enforce timetables or plan services.
- **Poor communication of disruptions.** Delays and cancellations are rarely communicated to passengers in advance.

4. Proposed Solution: RideTrack

RideTrack is a cloud-based platform made of a secure backend (REST API and real-time WebSocket service), a passenger mobile app, a staff ticket-scanning app and a web admin panel for authorities. Its main features are:

- **Live vehicle tracking:** GPS devices or phones on each vehicle send positions; passengers follow vehicles on a map with live arrival estimates compared with the timetable.
- **Timetables:** departures per weekday in the local time zone, automatically turned into daily trips.
- **Digital QR ticketing:** tickets are bought in the app and carry a cryptographically signed QR code that conductors scan; a manual ticket-ID check is available as a fallback.
- **Cashless payments:** a pluggable payment gateway with signed confirmations, so it can connect to local providers (for example LankaPay-connected banks or mobile wallets).
- **Automatic delay alerts:** the system detects late trips and notifies affected passengers by in-app message and optional push notification.
- **Occupancy reporting:** staff report how full a vehicle is so passengers and planners can see crowding.
- **Authority dashboard and reports:** live operations view, ticket sales, revenue and delay reports, and management of users, routes, stops, vehicles and trips.

5. Current Status of the Project

RideTrack is not only an idea: the backend is built, tested and ready to deploy. Current figures from the source code:

43	18	10	65	5
REST endpoints	database tables	API modules	automated tests	background jobs

- **Sri Lanka demo data:** Colombo bus routes 138 and 177 and the Main Line rail service to Kandy, with fares in LKR and timetables on Sri Lanka time.
- **Proven on real government data:** the platform has imported the official open GTFS timetable feed of Dubai's Roads and Transport Authority (16 Metro, tram and bus routes, 257 stops, real fare zones and timetables). The same import works for any GTFS feed, so Sri Lankan timetables can be loaded in the same way once available.
- **Simulator:** a built-in vehicle simulator drives buses along their routes, so the full system (live map, arrival times, delay alerts) can be demonstrated before any hardware is installed.
- **Deployment-ready:** runs in Docker containers on a cloud host, on MySQL or Microsoft SQL Server (including Azure SQL), so it can fit government data-centre or cloud policies.
- **Admin panel:** a web back office for authority officers is built into the system.

6. Technical Architecture and Security

Layer	Technology
Backend	Node.js 20, Express 4, REST API (/api/v1)
Real-time	Socket.IO WebSockets for live positions, alerts and dashboard updates
Database	MySQL 8 or Microsoft SQL Server 2017+ / Azure SQL
Mobile and web	Passenger and staff mobile app (React Native / Expo); built-in web admin panel
Notifications	In-app alerts; optional Firebase Cloud Messaging push notifications
Deployment	Docker and Docker Compose; any cloud or on-premise server

Security and data protection

- Role-based access for passengers, staff and authority officers, using short-lived access tokens and rotating refresh tokens; passwords stored only as secure hashes.
- Signed QR tickets that cannot be forged; payment confirmations verified with HMAC signatures.
- Protection against common web attacks: security headers, rate limiting, strict input validation.
- Separate keys for GPS devices; all secrets configurable per deployment.
- Designed to keep passenger data to the minimum needed, in line with Sri Lanka's Personal Data Protection Act, No. 9 of 2022.

7. Proposed Pilot and Implementation Plan

I propose a phased approach that starts small, measures results and scales only after success. Routes, vehicle numbers and timing below are suggestions to be agreed with the Task Force, NTC and SLTB.

Phase	Duration	Activities
0. Presentation and approval	Month 1	Live demonstration to the Task Force; agree pilot routes, partners, data access and success measures.
1. Preparation	Months 1-2	Load official timetables and stops for the pilot routes; set up hosting; connect a local payment provider in test mode; train conductors.
2. Pilot	Months 3-5	Run on 2-3 Colombo routes (for example 138 and 177) with about 20 SLTB buses using low-cost GPS units or phones; cash remains available alongside digital tickets.
3. Review	Month 6	Measure results against the agreed targets; collect passenger and staff feedback; publish a report to the Task Force.
4. Scale-up	Month 7 onwards	Extend to more Western Province routes, new low-floor buses, selected Sri Lanka Railways services, and then other provinces.

8. Expected Outcomes and Success Measures

Measure	How it is tracked	Pilot target (proposed)
On-time trips	GPS vs. timetable, delay reports	Measured baseline, then improvement
Digital ticket share	Tickets sold in app vs. total	At least 20% of pilot fares
Waiting time	Passenger survey and app usage	Shorter perceived waiting time
Fare transparency	Digital transactions per trip	Every digital fare traceable
Passenger satisfaction	In-app and on-board survey	Majority positive feedback
System reliability	Uptime monitoring	99% availability during service hours

9. Support Requested from the Government

- Endorsement of the pilot under the Clean Sri Lanka Programme.
- Coordination with the **NTC**, **SLTB** and **Sri Lanka Railways** for route, stop and timetable data and for pilot vehicles and conductors.
- Support to install GPS units (or staff phones) on pilot vehicles.
- Introduction to an approved payment provider for cashless fares.
- Hosting on government or approved cloud infrastructure, and modest funding for devices, hosting and development during the pilot.

Indicative cost items

Exact amounts will be prepared once the pilot scope is agreed. The main cost items are:

Item	Notes
GPS tracking units	One per pilot vehicle; staff smartphones can be used instead to cut cost
Cloud hosting and database	Small server and managed database for the pilot period
Payment gateway fees	Per-transaction fees charged by the payment provider
Development and support	Local adaptation (Sinhala and Tamil interface), integration and on-site support
Training and awareness	Conductor training; passenger awareness at pilot halts

10. Risks and Mitigation

Risk	Mitigation
Low smartphone or card use among some passengers	Keep cash available; allow conductors to issue tickets; add Sinhala and Tamil interfaces.
Resistance from operators or crews	Involve bus associations early; show how tracking and digital fares protect honest crews and owners.
Poor mobile coverage on some routes	Devices store positions and send them when back online; tickets can be checked by ID offline.
Data privacy concerns	Minimal personal data, encryption and access controls in line with the Personal Data Protection Act.
Cost of hardware	Start with staff phones; phased roll-out based on pilot results.

11. Conclusion

RideTrack offers a practical, already-built and affordable way to bring live information, digital ticketing and cashless payments to Sri Lanka's buses and trains. It supports the Clean Sri Lanka vision of a disciplined, transparent and people-friendly public service. I respectfully request the Task Force's approval to present the system and to begin a pilot with the relevant transport authorities.

Contact details

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